

Embedding various even cycles in a hypercube with node failures *

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Abstract

Let f_v denote the number of faulty vertices in an n -dimensional hypercube Q_n . In this paper, we prove that every fault-free edge of Q_n for $n \geq 4$ and every fault-free vertex of Q_n for $n \geq 3$ lies on a fault-free cycle of every even length from 6 to $2^n - 2f_v$ inclusive if $f_v = n - 1$, and all faulty vertices are not adjacent to the same vertex.

Keywords: interconnection network, hypercube, hamiltonian, fault-tolerant, cycle embedding.

1 Introduction

In this paper, fundamental graph terminologies and notations follow [1]. A bipartite graph is one whose vertex set can be partitioned into two subsets X and Y , so that each edge has one end-vertex in X and another in Y , such a partition $\{X, Y\}$ is called a bipartition of the graph. We often use the symbol $G = (X \cup Y, E)$ to denote a bipartite graph $G = (V, E)$ with a bipartition $\{X, Y\}$. Two vertices are adjacent when they are incident with a common edge. A path of length k from x to y is a finite sequence of adjacent vertices written as $\langle v_0, v_1, v_2, \dots, v_k \rangle$, where $v_0 = x$, $v_k = y$, and all the vertices $v_0, v_1, v_2, \dots, v_k$ are distinct except possibly $v_0 = v_k$. For convenience, we use the sequence $\langle v_0, \dots, v_i, P[v_i, v_j], v_j, \dots, v_m \rangle$ to denote the path $\langle v_0, v_1, v_2, \dots, v_k \rangle$, where $P[v_i, v_j] = \langle v_i, v_{i+1}, \dots, v_j \rangle$ and the two vertices v_i and v_j are called the *end-vertices* of $P[v_i, v_j]$. Sometimes, we also use P to denote a path $P[v_i, v_j]$. Let $l(P)$ denote the length of the path P that is the number of edges in P . In particular, P is a path of length zero if $l(P) = 0$. The *distance* between u and v in G is denoted by $d_G(u, v)$, which is the length of a

shortest path between u and v in G . Two paths are vertex-disjoint (also called disjoint) if and only if they have no vertices in common. A *cycle* C is a special path with at least three vertices such that the first vertex is the same as the last one. A cycle of length k is called a k -cycle; a k -cycle is *odd* or *even* depending on whether k is odd or even. A path (respectively, cycle) which traverses each vertex of G exactly once is *hamiltonian path* (respectively, *hamiltonian cycle*). For consistency, we always use $F = F_e \cup F_v$ to denote the faulty set of Q_n in the following discussion, where $F_e \subset E(G)$ and $F_v \subset V(G)$. $G - F$ denotes the subgraph of G induced by $\{x \mid x \in V(Q_n) \text{ and } x \notin F_v\}$. A vertex u is a *fault-free* vertex if u is not in F . An edge $e = (u, v)$ is a *fault-free* edge if u , v , and (u, v) are not in F . A cycle is *fault-free* if it contains neither a faulty vertex nor a faulty edge.

An n -dimensional hypercube (n -cube for short), denoted by Q_n , is an undirected graph which contains 2^n vertices and has n edges per vertex. The vertex set V of Q_n consists of all the n -bit binary strings, i.e., $V = \{u_{n-1}u_{n-2} \dots u_1u_0 \mid u_i \in \{0, 1\}, i = 0, 1, \dots, n-1\}$. Two vertices $u = u_{n-1}u_{n-2} \dots u_1u_0$ and $v = v_{n-1}v_{n-2} \dots v_1v_0$ are linked by an edge if and only if u and v exactly differ in one bit position. The *Hamming distance* $h(u, v)$ between two vertices u and v is the number of different bits in the corresponding strings of both vertices. Let $u = u_{n-1}u_{n-2} \dots u_1u_0$ be a vertex in Q_n . The i -dimensional neighbor of u is $u^{(i)} = u_{n-1}u_{n-2} \dots \bar{u}_i \dots u_1u_0$, where $\bar{u}_i$ represents the *one's complement* of u_i . An edge (u, v) in $E(Q_n)$ is of *dimension* i if $v = u^{(i)}$. For a given $0 \leq j < n$, we can partition Q_n along dimension j into two $(n-1)$ -cubes such that Q_{n-1}^0 denotes the subgraph of Q_n induced by $\{x \in V(Q_n) \mid x_j = 0\}$ and Q_{n-1}^1 denotes the subgraph of Q_n induced by $\{x \in V(Q_n) \mid x_j = 1\}$. We have that Q_{n-1}^0 and Q_{n-1}^1 are isomorphic to Q_{n-1} . The edge (a, b) is called the *crossing edge* if a is in one $(n-1)$ -cube

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and b is in the other $(n-1)$ -cube, respectively. The following properties of Q_n are useful in subsequent proofs.

Lemma 1 *Let u and v be two distinct vertices of Q_n . Then there is a partition which can partition Q_n into two $(n-1)$ -cubes such that one contains u and the other contains v .*

Lemma 2 [5] *Let $n \geq 3$ and $f_v \leq n-2$. Then, every fault-free edge of Q_n lies on a fault-free cycle of every even length from 4 to $2^n - 2f_v$ inclusive.*

Definition 1 *A vertex u is called **i -rescuable** if there are exactly i fault-free edges incident with u . A vertex u is called **isolated** if none of fault-free edge is incident with u .*

Let F be a faulty set of Q_n and u be a fault-free vertex in Q_n . We define $N(u) = \{i \mid (u, u^{(i)}) \text{ as being a fault-free edge for } 0 \leq i \leq n-1\}$ and $\bar{N}(u) = \{i \mid u^{(i)} \in F \text{ or } (u, u^{(i)}) \in F \text{ for } 0 \leq i \leq n-1\}$. Obviously, $N(u) \cap \bar{N}(u) = \emptyset$ and $N(u) \cup \bar{N}(u) = \{0, 1, 2, \dots, n-1\}$.

Definition 2 *A fault-free edge $(u, u^{(j)})$ is **critical** if $N(u) \cap N(u^{(j)}) = \{j\}$, i.e., if (u, v) is a critical edge and $v = u^{(j)}$, there is no integer i except j such that $(u, u^{(i)})$ and $(v, v^{(i)})$ are fault-free, respectively.*

In the following sections, our methods are based on a simple and efficient recursive construction. The main idea is to carefully partition an n -cube into two $(n-1)$ -cubes. Subsequently, we establish two small vertex-disjoint cycles in the two $(n-1)$ -cubes. Finally, we merge the two cycles to form a desired cycle. Throughout this paper, f_e denotes the number of faulty edges in Q_n and f_v denotes the number of faulty vertices in Q_n .

2 Cycle embedding when $f_v = n-1$

Let $F \subseteq V(Q_n)$ and $f_v = n-1$. If u is a 1-rescuable vertex, all faulty vertices are adjacent to u . In this section, we discuss a faulty Q_n that $f_v = n-1$ and all faulty vertices are not adjacent to the same vertex, i.e., every fault-free vertex is incident with at least two fault-free edges. Moreover, $|N(u)| \geq 2$ for all fault-free vertex u in Q_n . One can observe that $|\bar{N}(u) \cup \bar{N}(v)| = n-1$ if (u, v) is a critical edge of Q_n .

Lemma 3 *Let $F \subset V(Q_n)$ with $|F| = n-1$ and every fault-free vertex is adjacent to at least two*

fault-free vertices. Then, a fault-free edge e is not a critical edge of Q_n if and only if it lies on a fault-free 4-cycle of $Q_n - F$.

Proof. Let u and v be two end-vertices of a fault-free edge e and $v = u^{(j)}$ for some $0 \leq j \leq n-1$. Suppose that e is a critical edge. By definition of critical edge, $N(u) \cap N(v) = \{j\}$. Assume that there exists a fault-free 4-cycle C_4 containing the edge e . Hence $C_4 = \langle u, u^{(i)}, v^{(i)}, v, u \rangle$ for some $0 \leq i \neq j \leq n-1$. Consequently, $|N(u) \cap N(v)| \geq 2$. It contradicts the definition of critical edge. Therefore, none of 4-cycle is fault-free which contains the edge e . Suppose that e is not a critical edge. Hence $|N(u) \cap N(v)| \geq 2$. It is not difficult to show that there exists a fault-free 4-cycle containing the edge e . $\square$

Let $F = \{x, y\}$ be a faulty set of Q_3 . If $h(x, y) = 2$, there exists a vertex z such that x and y are adjacent to z , i.e., the vertex z becomes a 1-rescuable vertex. Hence we only discuss Q_3 that contains two faulty vertices x and y , and $h(x, y) = 1$ and $h(x, y) = 3$. Since Q_n is vertex-symmetric for all $n \geq 2$, we may assume that $x = 000$, and thus, $y \in \{001, 010, 100\}$ if $h(x, y) = 1$ or $y = 111$ if $h(x, y) = 3$. Since Q_n is edge-symmetric, we only consider the case of that $y = 001$ and $y = 111$. When $F = \{000, 001\}$, $Q_3 - F$ is a graph such as an illustration of Figure 1 (a). It is not difficult to show that every edge of $Q_3 - F$ lies a fault-free 4-cycle. When $F = \{000, 111\}$, $Q_3 - F$ is a 6-cycle, as illustrated in Figure 1 (b). Consequently, every edge of $Q_3 - F$ lies on a fault-free 6-cycle. Therefore, Lemma 4 is true.

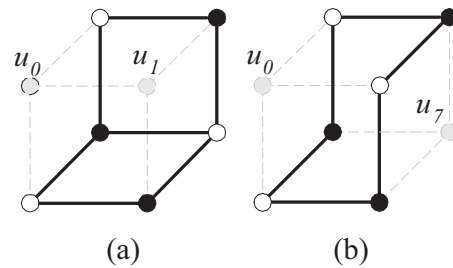


Figure 1: (a) An illustration of case of $F = \{u_0, u_1\}$ where $u_0 = 000$ and $u_1 = 001$. (b) An illustration of case of $F = \{u_0, u_7\}$ where $u_0 = 000$ and $u_7 = 111$.

Lemma 4 Let $f_e = 0$, $f_v = 2$ and every fault-free vertex is adjacent to at least two fault-free vertices in Q_3 . Then,

- (1) every fault-free edge of Q_3 lies on a fault-free 4-cycle, or
- (2) every fault-free edge of Q_3 lies on a fault-free 6-cycle.

A 4-cube is a basis case of this section. Let w_1 , w_2 , and w_3 be three faulty vertices of Q_4 . Suppose that these three faulty vertices are in the same partite set. One can observe that Q_4 has exactly four 2-rescuable vertices if these three faulty vertices are not adjacent to the same vertex. Since 4-cube is a bipartite, Q_4 has at most two vertices to be 2-rescuable if these three faulty vertices are distributed into two partite sets. Particularly, it is not difficult to show Lemma 5.

Lemma 5 Let w_1 , w_2 , and w_3 be three faulty vertices of Q_4 and they are not adjacent to the same vertex. Then,

- (1) Q_4 has exactly four 2-rescuable vertices if and only if these three faulty vertices are in the same partite set.
- (2) Q_4 has at most two vertices, u and v , to be 2-rescuable, and u and v are adjacent to two faulty vertices in common if w_1 , w_2 , and w_3 are distributed into two partite sets.

Let $f_e = 0$, $f_v = n - 1$ and every fault-free vertex is adjacent to at least two fault-free vertices in Q_n for $n \geq 4$. Using the simple arithmetic, one may observe the following results. (1) A Q_5 has at most two vertices, u and v , to be 2-rescuable, and u and v are adjacent to two faulty vertices in common; (2) For $n \geq 6$, Q_n has at most one 2-rescuable vertex; (3) For $n \geq 5$, there is at most one critical edge in Q_n . Lemma 6 is easy to prove.

Lemma 6 Let $f_e = 0$, $f_v = n - 1$ and every fault-free vertex is adjacent to at least two fault-free vertices in Q_n for $n \geq 5$. Suppose that (a, b) is a critical edge. Then,

- (1) $|N(a)| = 2$ or $|N(b)| = 2$ if there exists a 2-rescuable vertex in Q_n .
- (2) $|N(u)| \geq 3$ for all fault-free vertex u in Q_n if $|N(a)| \geq 3$ and $|N(b)| \geq 3$.

We partition Q_n along dimension j according to the following methods in the proofs of Lemma 8 and Theorem 1.

- (1) Suppose that Q_n contains exactly two vertices, u and v , to be 2-rescuable. Hence u and

v are adjacent to two faulty vertices $u^{(j)}$ and $u^{(k)}$ for some $0 \leq j, k \leq n - 1$. One may partition Q_n along dimension j into two $(n - 1)$ -cubes, denoted by Q_{n-1}^0 and Q_{n-1}^1 . Consequently, $u^{(j)}$ and $u^{(k)}$ are in different $(n - 1)$ -cubes and every vertex of Q_{n-1}^0 (respectively, Q_{n-1}^1) is adjacent to at least two fault-free neighbors in Q_{n-1}^0 (respectively, Q_{n-1}^1).

- (2) Suppose that Q_n contains exactly one vertex u to be 2-rescuable. Let $u^{(i)}$ and $u^{(k)}$ be two fault-free neighbors of u . Choosing any integer j with $0 \leq j \leq n - 1$ such that $j \neq i$ and $j \neq k$, one may partition Q_n along dimension j into two $(n - 1)$ -cubes, denoted by Q_{n-1}^0 and Q_{n-1}^1 , such that each $(n - 1)$ -cube contains at least one faulty vertex. Furthermore, every vertex of Q_{n-1}^0 (respectively, Q_{n-1}^1) is adjacent to at least two fault-free neighbors in Q_{n-1}^0 (respectively, Q_{n-1}^1).
- (3) Suppose that Q_n contains exactly one critical edge (u, v) and none of 2-rescuable vertex exists. In this situation, u and v have at least one faulty neighbor, respectively, i.e., $|\overline{N}(u)| \geq 1$ and $|\overline{N}(v)| \geq 1$. Without loss of generality, we may assume $|\overline{N}(u)| \geq |\overline{N}(v)|$. Since $n \geq 4$, $|\overline{N}(u)| \geq 2$. Let $u^{(j)}$ be any faulty neighbor of u . One may partition Q_n along dimension j into two $(n - 1)$ -cubes, denoted by Q_{n-1}^0 and Q_{n-1}^1 , such that one contains the faulty node $u^{(j)}$ and the other contains the edge (u, v) and at least one faulty neighbor of u . Thus, every vertex of Q_{n-1}^0 (respectively, Q_{n-1}^1) is adjacent to at least two fault-free neighbors in Q_{n-1}^0 (respectively, Q_{n-1}^1).
- (4) Suppose that Q_n has neither 2-rescuable vertex nor critical edge. In this situation, every fault-free vertex of Q_n is adjacent to at least three fault-free vertices and every edge is non-critical. Applying Lemma 1, Q_n can be partitioned along some dimension j into two $(n - 1)$ -cubes such that each $(n - 1)$ -cube contains at least one faulty vertex.

Lemma 7 Let $n \geq 4$. Assume that Q_n is partitioned into two $(n - 1)$ -cubes by the previous method. Then, every crossing edge is non-critical.

We prove the main result of this section by an inductive schema. Lemma 8 is the induction base.

Lemma 8 *Let $f_e = 0$, $f_v = 3$, and every fault-free vertex is adjacent to at least two fault-free vertices in Q_4 . Then, every fault-free edge lies on a fault-free 6-cycle, 8-cycle, and 10-cycle in Q_4 .*

Proof. Let w_1 , w_2 , and w_3 be three faulty vertices of Q_4 and these three faulty vertices are not adjacent to the same vertex. Suppose that these three faulty vertices are in the same partite set. By Lemma 5, Q_4 has exactly four vertices to be 2-rescuable. It is not different to show this case by computer program. In the following proof, we only consider the case of that w_1 , w_2 , and w_3 are distributed into two partite sets, i.e., Q_4 has at most two vertices to be 2-rescuable.

Applying the previous methods, Q_4 can be partitioned along dimension j into two 3-cubes, denoted by Q_3^0 and Q_3^1 , such that each contains at least one faulty vertex and every vertex of Q_3^0 (respectively, Q_3^1) is adjacent to at least two fault-free neighbors in Q_3^0 (respectively, Q_3^1). Let F_i be the faulty vertex set of Q_{n-1}^i and $f_v^i = |F_i|$ for $i = 0, 1$. Thus, $1 \leq f_v^0 \leq 2$, $1 \leq f_v^1 \leq 2$, and $f_v^0 + f_v^1 = 3$. Let e be any fault-free edge of Q_n . In order to prove this lemma, we will establish a fault-free 6-cycle, 8-cycle, and 10-cycle such that the edge e is covered by them. We check two possible distributions of the fault-free edge e .

Case 1. $e \in E(Q_3^0) \cup E(Q_3^1)$, i.e., the edge e is in a 3-cube, Q_3^0 or Q_3^1 . We consider that $e \in E(Q_3^0)$ (the discussion of $e \in E(Q_3^1)$ is the same).

Subcase 1-1. $f_v^0 = 1$ and $f_v^1 = 2$. Since $f_v^0 = 1$ and by Theorem 2, the edge e lies on a fault-free 4-cycle and 6-cycle of Q_3^0 . Let C_6 be a fault-free 6-cycle in Q_3^0 including the edge e . Hence C_6 is the desired 6-cycle. Obviously, there are three mutually disjoint edges excluding e lying on C_6 . Since $f_v^1 = 2$, there exists an edge $(a, b) \neq e$ in C_6 such that $a^{(j)}$ (respectively, $b^{(j)}$) is a fault-free neighbor of a (respectively, b) in Q_3^1 . Thus, combining the cycle C_6 , and edges $(a, a^{(j)})$, $(b, b^{(j)})$, and $(a^{(j)}, b^{(j)})$, the desired 8-cycle is constructed.

Since every vertex of Q_3^1 has at least two fault-free neighbors in Q_3^1 and by Lemma 4, either every fault-free edge of $Q_3^1 - F_1$ lies on a 4-cycle or every fault-free edge of $Q_3^1 - F_1$ lies on a 6-cycle. We next construct a fault-free 10-cycle containing the edge e .

(a). Every fault-free edge of $Q_3^1 - F_1$ lies on a 4-cycle. Let D_4 be a fault-free 4-cycle including $(a^{(j)}, b^{(j)})$ in Q_3^1 . Thus, the desired 10-cycle can be obtained by combining the edge $(a, a^{(j)})$, the edge $(b, b^{(j)})$, the cycle C_6 , and the cycle D_4 .

(b). Every fault-free edge of $Q_3^1 - F_1$ lies on a 6-cycle. In this situation, the reduce graph $Q_3^1 - F_1$ is a 6-cycle, denoted by D_6 . Let $F_1 = \{x, y\}$. Obviously, $h(x, y) = 3$. This imply $h(x^{(j)}, y^{(j)}) = 3$. Let C_4 be a fault-free 4-cycle containing the edge e in Q_3^0 . Since $h(x^{(j)}, y^{(j)}) = 3$, C_4 passes through at most one vertex in $\{x^{(j)}, y^{(j)}\}$. Hence there exists an edge $(c, d) \neq e$ in C_4 such that $c, d \notin \{x^{(j)}, y^{(j)}\}$. Consequently, $(c^{(j)}, d^{(j)})$ is a fault-free edge in Q_3^1 . The desired 10-cycle can be constructed by merging the two cycles C_4 and D_6 as well as the two edges $(c, c^{(j)})$ and $(d, d^{(j)})$.

Subcase 1-2. $f_v^0 = 2$ and $f_v^1 = 1$. There are only two situations to be considered.

(a). Every fault-free edge of $Q_3^0 - F_0$ lies on a 4-cycle. Let C_4 be a fault-free 4-cycle passing the edge e in Q_3^0 . There are two disjoint edges excluding e lying on C_4 . This imply that a fault-free edge $(a, b) \neq e$ is existent in C_4 such that $a^{(j)}$ and $b^{(j)}$ are fault-free in Q_3^1 . Since $f_v^1 = 1$ and by Theorem 2, let D_4 (respectively, D_6) be a 4-cycle (respectively, 6-cycle) containing the edge $(a^{(j)}, b^{(j)})$ in Q_3^1 . Thus, the desired 6-cycle is constructed by merging C_4 , $(a, a^{(j)})$, $(b, b^{(j)})$, and $(a^{(j)}, b^{(j)})$. Finally, the desired 8-cycle (respectively, 10-cycle) is obtained by combining the two edges $(a, a^{(j)})$ and $(b, b^{(j)})$ as well as the cycle C_4 and the cycle D_4 (respectively, D_6).

(b). Every fault-free edge of $Q_3^0 - F_0$ lies on a 6-cycle. In this situation, the reduce graph $Q_3^0 - F_0$ is a 6-cycle, denoted by C_6 . Thus, C_6 is the desired 6-cycle. There are three mutually disjoint edges excluding e contained in C_6 . This imply that a fault-free edge $(c, d) \neq e$ is existent in C_6 such that $c^{(j)}$ and $d^{(j)}$ are fault-free in Q_3^1 . Since $f_1 = 1$ and by Theorem 2, $(c^{(j)}, d^{(j)})$ lies on a 4-cycle in Q_3^1 , denoted by D_4 . Therefore, the desired 8-cycle is obtained by combining the cycle C_6 , $(c, c^{(j)})$, $(d, d^{(j)})$, and $(c^{(j)}, d^{(j)})$. The desired 10-cycle is created by merging the two edges $(c, c^{(j)})$ and $(d, d^{(j)})$ as well as the cycle C_6 and D_4 .

Case 2. $e \notin E(Q_3^0) \cup E(Q_3^1)$, i.e., the edge e is a crossing edge between Q_3^0 and Q_3^1 . Let u and v be two end-vertices of e . Hence, $v = u^{(j)}$. Without loss of generality, we may assume that $u \in V(Q_3^0)$ and $v \in V(Q_3^1)$. According to previous partition methods, we have that any crossing edge is not a critical edge. Hence, $\langle u, u^{(r)}, v^{(r)}, v, u \rangle$ forms a fault-free 4-cycle for some $0 \leq r \neq j \leq 3$. Obviously, $u, u^{(r)} \in V(Q_3^0)$ and $v, v^{(r)} \in V(Q_3^1)$. Without loss of generality, we may assume that $f_0 = 1$. By Theorem 2, let C_4 (respectively, C_6) be a 4-cycle (respectively, 6-cycle) containing the edge $(u, u^{(r)})$ in Q_3^0 . The desired 6-cycle (respec-

tively, 8-cycle) is established by merging the 4-cycle $\langle u, u^{(r)}, v^{(r)}, v, u \rangle$ and C_4 (respectively, C_6).

We now construct the desired 10-cycle. Since $f_v^1 = 2$, there are two situations to be considered. One case is that every fault-free edge of $Q_3^1 - F_1$ lies on a 4-cycle. Let D_4 be a fault-free 4-cycle covering $(v, v^{(r)})$ in Q_3^1 . Merging the two edges (u, v) and $(u^{(r)}, v^{(r)})$ as well as the two cycles D_4 and C_6 , the desired 10-cycle can be constructed. The other case is that every fault-free edge of $Q_3^1 - F_1$ lies on a 6-cycle. Let D_6 be a fault-free 6-cycle passing $(v, v^{(r)})$ in Q_3^1 . Merging the two edges (u, v) and $(u^{(r)}, v^{(r)})$ as well as the two cycles D_6 and C_4 , the desired 10-cycle can be obtained. $\square$

Theorem 1 *Let $f_e = 0$, $f_v = n - 1$, and every fault-free vertex is adjacent to at least two fault-free vertices in Q_n for $n \geq 4$. Then, every fault-free edge of Q_n lies on a fault-free cycle of every even length from 6 to $2^n - 2n + 2$ inclusive.*

Proof. The following proof is induction on n . By Lemma 8, the theorem is true if $n = 4$. Assume that the result is true for $4 \leq m < n$. We now consider that $m = n$. Applying partition methods introducing to this section, one may partition Q_n along dimension j into two $(n - 1)$ -cubes, denoted by Q_{n-1}^0 and Q_{n-1}^1 , such that each contains at least one faulty vertex and every vertex of Q_{n-1}^0 (respectively, Q_{n-1}^1) is adjacent to at least two fault-free neighbors in Q_{n-1}^0 (respectively, Q_{n-1}^1). Let f_v^i denote the number of faulty vertices in Q_{n-1}^i for $i = 0, 1$. Thus, $1 \leq f_v^0 \leq n - 2$, $1 \leq f_v^1 \leq n - 2$ and $f_v^0 + f_v^1 = n - 1$. Let $e = (u, v)$ be any fault-free edge of Q_n . We will establish a cycle of every even length from 6 to $2^n - 2n + 2$ inclusive such that the edge e is included. We check two possible distributions of the fault-free edge e as following two cases.

Case 1. $e \in E(Q_{n-1}^0) \cup E(Q_{n-1}^1)$, i.e., the edge e is in an $(n - 1)$ -cube, Q_{n-1}^0 or Q_{n-1}^1 . We consider that $e \in E(Q_{n-1}^0)$ (the discussion of $e \in E(Q_{n-1}^1)$ is the same). Since $f_v^0 \leq n - 2$, by induction hypothesis, e lies on an even l -cycle in Q_{n-1}^0 where $6 \leq l \leq 2^{n-1} - 2f_v^0$. Thus, the desired even cycle of length from 6 to $2^{n-1} - 2f_v^0$ inclusive is obtained.

Subsequently, a cycle of every even length from $2^{n-1} - 2f_v^0 + 4$ to $2^n - 2n$ inclusive is constructed. Let C_k be a fault-free k -cycle covering the edge e in Q_{n-1}^0 where $k = 2^{n-1} - 2f_v^0 - 2$. Obviously, there are $2^{n-2} - f_v^0 - 1$ mutually disjoint edges excluding e in C_k . Since $2^{n-2} > n$ for $n \geq 5$, $2^{n-2} > f_v^0 + f_v^1 + 1$. This imply $2^{n-2} - f_v^0 - 1 > f_v^1$. Thus, there exists an edge $(a, b) \neq e$ in C_k such that $a^{(j)}$ (respectively, $b^{(j)}$) is a fault-free neighbor

of a (respectively, b) in Q_{n-1}^1 . Hence C_k may be formed by $\langle a, R_0[a, b], b, a \rangle$ where $l(R_0) = 2^{n-1} - 2f_v^0 - 3$ and e is included in the path R_0 . Since $f_v^1 \leq n - 2$, by induction hypothesis, there is an even h cycle of $D_h = \langle b^{(j)}, R_1[b^{(j)}, a^{(j)}], a^{(j)}, b^{(j)} \rangle$ covering the edge $(a^{(j)}, b^{(j)})$ in Q_{n-1}^1 where $6 \leq h \leq 2^{n-1} - 2f_v^1$. Obviously, $l(R_1) = 5, 7, 9, \dots, 2^{n-1} - 2f_v^1 - 1$. Combining the two cycles C_k and D_h as well as the two edges $(a, a^{(j)})$ and $(b, b^{(j)})$, we have an even g -cycle of $C_g = \langle a, R_0[a, b], b, b^{(j)}, R_1[b^{(j)}, a^{(j)}], a^{(j)}, a \rangle$ covering the edge e where $g = l(R_0) + l(R_1) + 2$. Therefore, $2^{n-1} - 2f_v^0 + 4 \leq g \leq 2^n - 2n$.

We now construct the desired two cycles of lengths $2^{n-1} - 2f_v^0 + 2$ and $2^n - 2n + 2$. Let C_r be an even r -cycle containing the edge e with $r = 2^{n-1} - 2f_v^0$ in Q_{n-1}^0 . Similarly, we choose an edge $(c, d) \neq e$ in C_r such that $c^{(j)}$ (respectively, $d^{(j)}$) is a fault-free neighbor of c (respectively, d) in Q_{n-1}^1 . Combining the cycle C_r , $(c, c^{(j)})$, $(d, d^{(j)})$, and $(c^{(j)}, d^{(j)})$, a cycle of length $2^{n-1} - 2f_v^0 + 2$ is constructed and it contains the edge e . Let D_s be a fault-free even s -cycle with $s = 2^{n-1} - 2f_v^1$ in Q_{n-1}^1 passing through the edge $(c^{(j)}, d^{(j)})$. Merging the two cycles C_r and D_s as well as the two edges $(c, c^{(j)})$ and $(d, d^{(j)})$, the desired $(2^n - 2n + 2)$ -cycle is constructed.

Case 2. $e \notin E(Q_{n-1}^0) \cup E(Q_{n-1}^1)$, i.e., the edge e is a crossing edge between Q_{n-1}^0 and Q_{n-1}^1 . Let u and v be two end-vertices of e . Without loss of generality, we may assume that $u \in V(Q_{n-1}^0)$ and $v \in V(Q_{n-1}^1)$. Hence e is non-critical. Specifically, $|N(u) \cap N(v)| \geq 2$, i.e., there exists an integer $r (\neq j)$ such that $u^{(r)}$ (respectively, $v^{(r)}$) is a fault-free neighbor of u (respectively, v). Thus, $\langle u, u^{(r)}, v^{(r)}, v, u \rangle$ forms a fault-free 4-cycle, and $u, u^{(r)} \in V(Q_{n-1}^0)$ and $v, v^{(r)} \in V(Q_{n-1}^1)$. Since $f_v^0 + f_v^1 = n - 1$ and $n \geq 5$, either $f_v^0 \leq n - 3$ or $f_v^1 \leq n - 3$. Without loss of generality, we may assume that $f_v^0 \leq n - 3$. By Theorem 2, the edge $(u, u^{(r)})$ lies on a fault-free even k -cycle C_k in Q_{n-1}^0 where $4 \leq k \leq 2^{n-1} - 2f_v^0$. Since $f_v^1 \leq n - 2$, by induction hypothesis, the edge $(v, v^{(r)})$ lies on a fault-free even h -cycle D_h in Q_{n-1}^1 where $6 \leq h \leq 2^{n-1} - 2f_v^1$.

Merging the 4-cycle $\langle u, u^{(r)}, v^{(r)}, v, u \rangle$ and a 4-cycle (respectively, 6-cycle) including $(u, u^{(r)})$, the desired 6-cycle (respectively, 8-cycle) is constructed. Subsequently, combining the two cycles C_k and D_h as well as the two edges (u, v) and $(u^{(r)}, v^{(r)})$, the desired even l -cycle H_l is obtained where $10 \leq l \leq 2^n - 2n + 2$. $\square$

Since $f_e = 0$, $f_v = n - 1$, and all faulty vertices are not adjacent to the same vertex, every fault-free vertex of Q_n has at least two fault-free neighbors. When $n = 3$, $f_e = 0$, and $f_v = 2$, it is not difficult to show that every fault-free vertex in Q_3 lies on a 6-cycle. Therefore, Corollary 1 is true.

Corollary 1 *Let $f_e = 0$, $f_v = n - 1$, and every fault-free vertex is adjacent to at least two fault-free vertices in Q_n for $n \geq 3$. Then, every fault-free vertex of Q_n lies on a fault-free cycle of every even length from 6 to $2^n - 2n + 2$ inclusive.*

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